

5 The number and size of Golgi apparatus vary depending on the type of cell.

(a) (i) Draw a labelled diagram of the Golgi apparatus.

(4)

(ii) How many of the following are functions of the Golgi apparatus?

- formation of extracellular enzymes
- modification of proteins
- formation of peptide bonds through condensation reactions

(1)

☐ **A** none

☐ **B** one

☐ **C** two

☐ **D** three

(iii) A cell was supplied with radioactive amino acids. The cell took in these amino acids and used them in protein synthesis.

Which structure in the cell would become radioactive first?

(1)

☐ **A** centrioles

☐ **B** Golgi apparatus

☐ **C** lysosomes

☐ **D** ribosomes



(b) In a mammalian cell, there was one Golgi apparatus with a diameter of $1.1\text{ }\mu\text{m}$.

In a maize cell, there were 600 Golgi apparatus. The largest diameter of Golgi apparatus in the maize cell was $0.5\text{ }\mu\text{m}$.

(i) Calculate the percentage difference in these diameters.

(1)

Answer%

(ii) The number and size of the Golgi apparatus in the maize cell increases during one stage of the cell cycle.

In which stage of the cell cycle would the number and size of the Golgi apparatus increase?

(1)

- ☐ **A** anaphase
- ☐ **B** interphase
- ☐ **C** prophase
- ☐ **D** telophase



(iii) Suggest why the number and size of the Golgi apparatus change during the cell cycle.

(4)

(Total for Question 5 = 12 marks)

